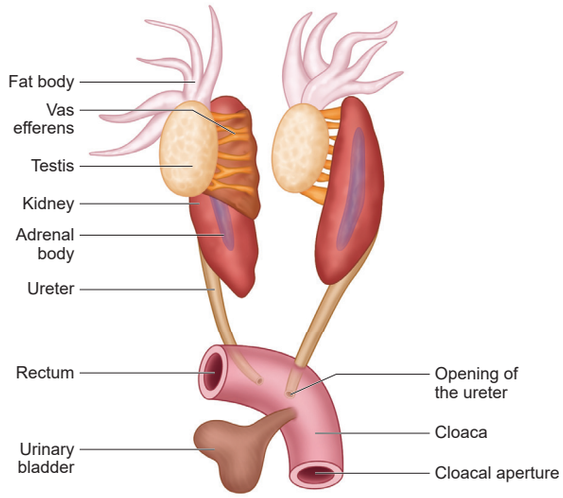


Figure 4.23 *Rana hexadactyla* - Male Urinogenital System

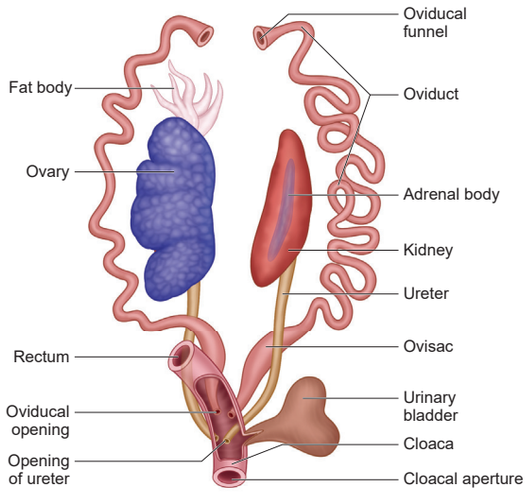


Figure 4.24 *Rana hexadactyla* - Female Urinogenital System

Excretory system

Elimination of **nitrogenous waste** and salt and water balance are performed by a well developed excretory system. It consists of a pair of kidneys, ureters, urinary bladder and cloaca. Kidneys are dark red, long, flat organs situated on either sides of the vertebral column in the body cavity. Kidneys are **Mesonephric.** Several nephrons are found in each kidney. They separate nitrogenous waste from the blood and excrete urea, so frogs are called **ureotelic** organisms. A pair of ureters emerges from the kidneys and opens into the cloaca. A thin walled unpaired **urinary bladder** is present ventral to the **rectum** and opens into the **cloaca.**

Reproductive system

The **male** frog has a pair of testes which are attached to the kidney and the dorsal body wall by folds of peritonium called mesorchium. Vasa efferentia arise from each **testis**. They enter the kidneys on both side and open into the bidder's canal. Finally, it communicates with the urinogenital duct that comes out of kidneys and opens into the cloaca (Figure 4.23).

Female reproductive system (Figure 4.24) consists of paired **ovaries,** attached to the kidneys, and dorsal body wall by folds of peritoneum called mesovarium. There is a pair of coiled **oviducts** lying on the sides of the kidney. Each oviduct opens into the body-cavity at the anterior end by a funnel like opening called ostia. Unlike the male frog, the female frog has separate genital ducts distinct from ureters. Posteriorly the oviducts dilated to form **ovisacs** before they open into cloaca. Ovisacs store the eggs temporarily before they are sent out through the cloaca. Fertilization is external.

Within few days of fertilization, the **eggs** hatch into **tadpoles**. A newly hatched tadpole lives off the yolk stored in its body. It gradually grows larger and develops three pairs of gills. The tadpole grows and **metamorphosis** into an air – breathing carnivorous adult frog (Figure 4.25). Legs grow from the body, and the tail and gills disappear. The mouth broadens, developing teeth and jaws, and the lungs become functional.

XI Std Zoology Chapter-4 EM.indd 68 30/12/2022 10:37:04